

Beyond Accuracy: Why Ensembles Work... Until They Don't!

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Motivation

Ensemble methods are widely regarded as reliable performance boosters. Adding more models and averaging their predictions often improves accuracy, but this improvement is typically evaluated only through aggregate error metrics. The problem is that accuracy, as a single number, provides no insight into the underlying error dynamics. It fails to show whether error reduction stems from **decreased variance, reduced bias, or both**; furthermore, it cannot explain why combining methods can produce results neither achieves alone. While **bias-variance decomposition** provides a classical theoretical framework for understanding generalization in regression, it is rarely used to compare ensemble families systematically. **This project investigates ensemble behavior through explicit bias-variance geometry**, revealing how different methods move through this space and under what conditions hybrid constructions can outperform their components.

Research Question

We ask: how do ensemble methods traverse bias-variance space, and what does this reveal about the mechanisms driving their performance that accuracy alone cannot reveal? Specifically:

- How do **bagging and boosting** trace distinct trajectories in the bias-variance plane as ensemble size, base complexity, dataset size, and noise vary?
- Do hybrid constructions **interpolate** between their parents, or can they reach **strictly better bias-variance tradeoffs**?
- Can **covariance** between parent predictors predict when **convex mixtures** will improve total error?
- Does **diversity** remain a reliable indicator of ensemble performance under **noise**, or does noise-induced disagreement become a misleading signal?

Approach and Algorithms

We work in **regression under squared loss**, allowing exact **bias-variance decomposition** with empirical estimation via repeated bootstrap resampling. Each model configuration is represented as a point in a 2D plane with axes **Bias² and Variance**. Varying ensemble parameters produces trajectories in this space, enabling structural comparison across methods. Plotting trajectories rather than a single accuracy number reveals the path a method takes, not just where it ends up, making it a richer and more diagnostic representation of ensemble behavior.

We study three homogeneous, tree-based ensemble families:

- Random Forest, as a representative of bagging
- XGBoost, as a representative of boosting
- Two hybrid constructions:
 - **Architectural hybrid**: bagging applied to boosted models (BaggingRegressor wrapping XGBoost), combining both mechanisms at training time.
 - **Convex combination hybrid**: a post-hoc mixture $f_\lambda = \lambda \cdot f_{\text{boost}} + (1-\lambda) \cdot f_{\text{bag}}$ swept across $\lambda \in [0,1]$. We derive closed-form expressions for Bias² and Variance as functions of λ , incorporating the covariance between parent predictors, yielding an analytic optimal coefficient λ^* , which we compare against empirical results.

Experiments and Analysis

For each configuration, models are trained across $M=50$ independent resampled datasets, enabling empirical estimation of Bias² and Variance. For hybrid analysis, covariance between parent predictors is estimated using shared bootstrap samples across both models, ensuring predictions are comparable trial by trial. We vary one parameter at a time relative to a fixed baseline:

- Dataset noise level
- Training set size
- Base learner complexity
- Ensemble size

Diversity is defined via the ambiguity decomposition as the difference between average individual model variance and ensemble variance: $(1/k) \sum \text{Var}(f_i) - \text{Var}(\bar{f})$.

This quantifies variance reduction attributable to disagreement among ensemble members. We evaluate whether diversity remains a reliable predictor of ensemble performance under increasing noise, and whether covariance between parent models predicts the curvature and optimality of convex mixtures.

Datasets

Primary experiments use **synthetic** regression data generated via scikit-learn's `make_regression`, with additional nonlinear targets (e.g., sinusoidal functions) to ensure bias is non-trivial and boosting has meaningful bias-reduction capacity. These datasets allow controlled variation of sample size, feature dimensionality, number of informative features, and label noise, enabling clean isolation of each parameter's effect on bias-variance trajectories.

We validate all findings on the **California Housing dataset** (~20,000 samples, 8 features) to assess whether observed bias-variance behaviors persist when noise is real, uncontrolled, and feature interactions are not artificially constructed. This two-tier design ensures findings are theoretically grounded on synthetic data and practically relevant on real data.

Plan and Milestones

By the pre-final check-in, we aim to complete:

- All synthetic experiments across the four parameter axes
- Derivation of the convex mixture bias-variance expressions and optimal λ^*
- Empirical covariance estimation and diversity analysis
- Full bias-variance trajectory visualizations across all conditions

Final validation on California Housing and synthesis of theoretical-empirical comparisons will follow before the submission deadline.